

Physical Chemistry (II)

Lecture 02

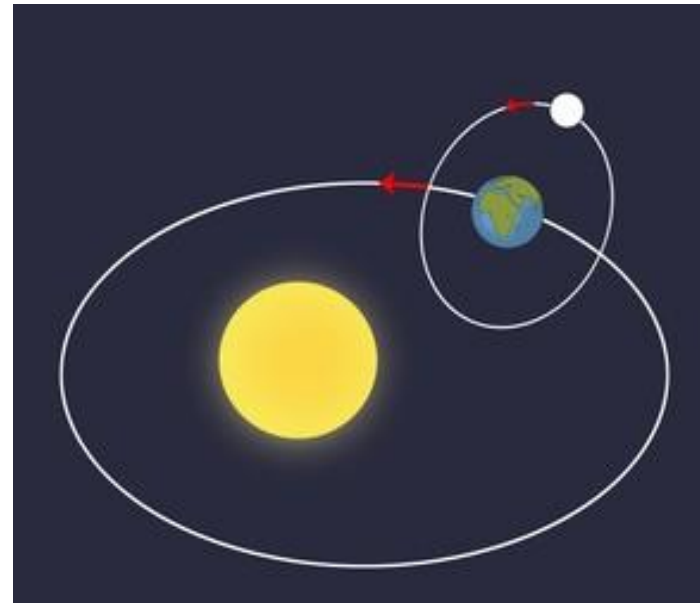
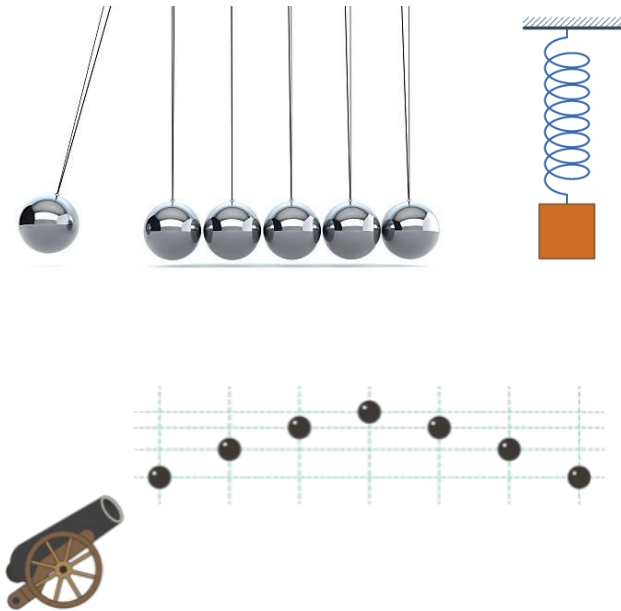
Particles, Waves, and Duality



RS 1.3

Classical Mechanics

Physical theory describing the motion of macroscopic objects



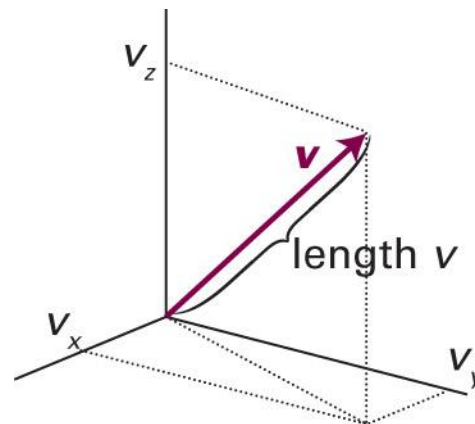
Physics of PARTICLES

Position and Velocity

Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$



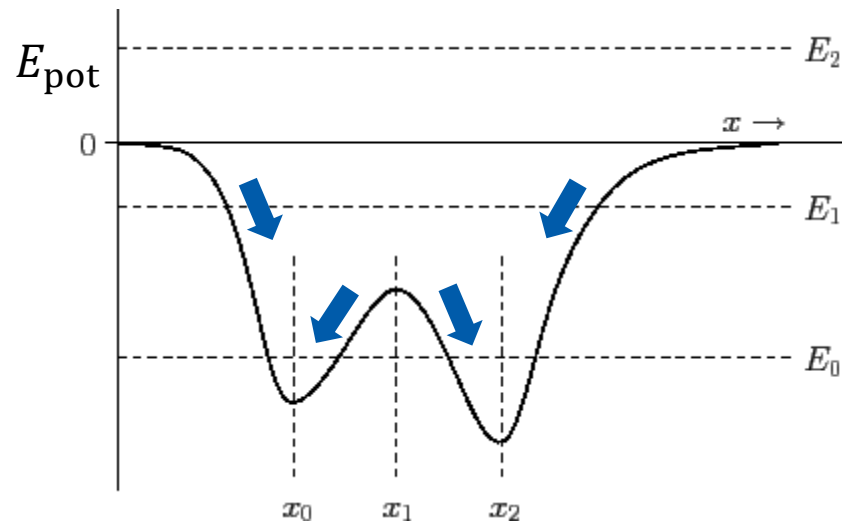
With the velocity, we can define:

- Momentum $\vec{p} = m\vec{v}$
- Kinetic energy $T = mv^2/2 = m(v_x^2 + v_y^2 + v_z^2)/2$

Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Force $\vec{F}$: negative slope of potential energy ($-\nabla V$)

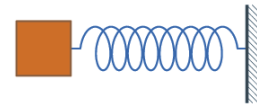


(Image: <http://farside.ph.utexas.edu/teaching/336k/Newtonhtml/node16.html>)

Examples of Potential Energy

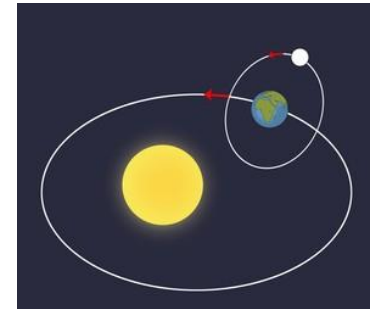
- Elastic potential energy (“spring”)

$$V(x) = \frac{1}{2} k(x - x_0)^2$$



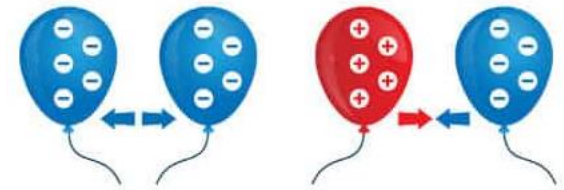
- Gravitational potential energy

$$V(r) = -G \frac{m_1 m_2}{r}$$



- Electrostatic potential energy

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$



Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Acceleration $\vec{a}$: second time-derivative of position

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

$$-\nabla V = m \frac{\partial^2 \vec{r}}{\partial t^2}$$

↑
given

Second-Order ODE

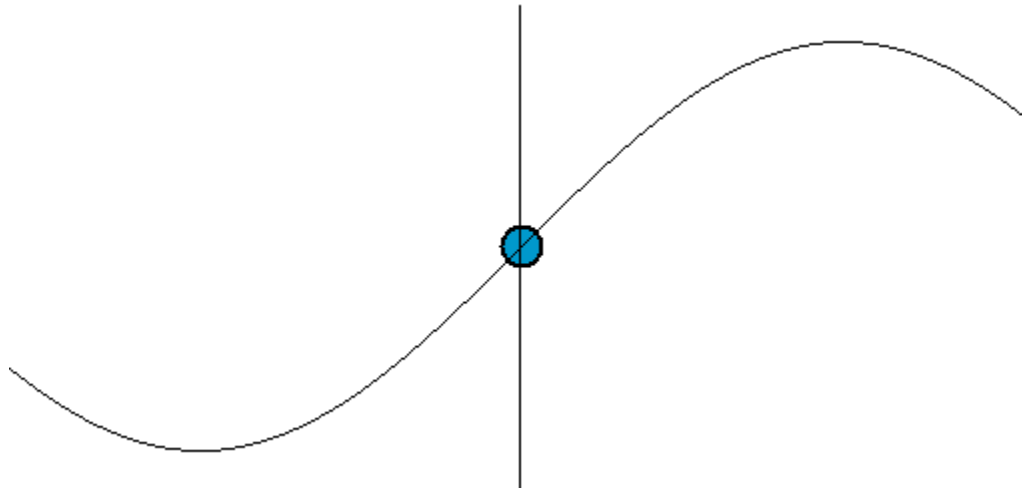
$$\frac{\partial^2 \vec{r}}{\partial t^2} = -\frac{\nabla V}{m}$$

Two conditions are required to obtain the full dynamics:

(initial) position and (initial) velocity (= momentum)

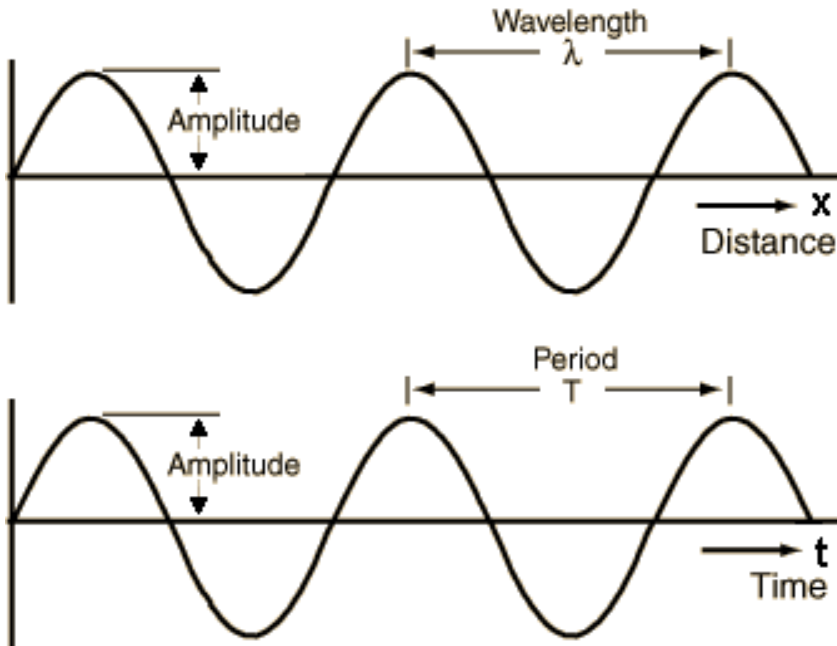
Waves

Temporal change + Spatial change



(Image: https://commons.wikimedia.org/wiki/File:Simple_harmonic_motion_animation.gif)

Basic Terminology



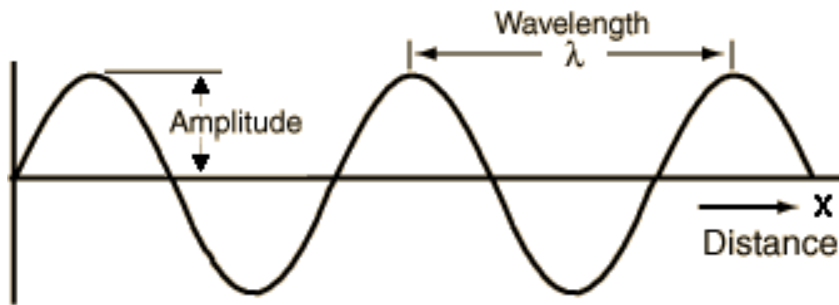
$$\text{Wavenumber } \tilde{\nu} = 1/\lambda$$

$$\text{Frequency } \nu = 1/T$$

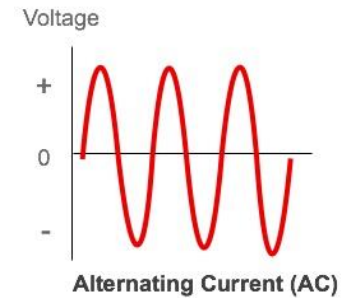
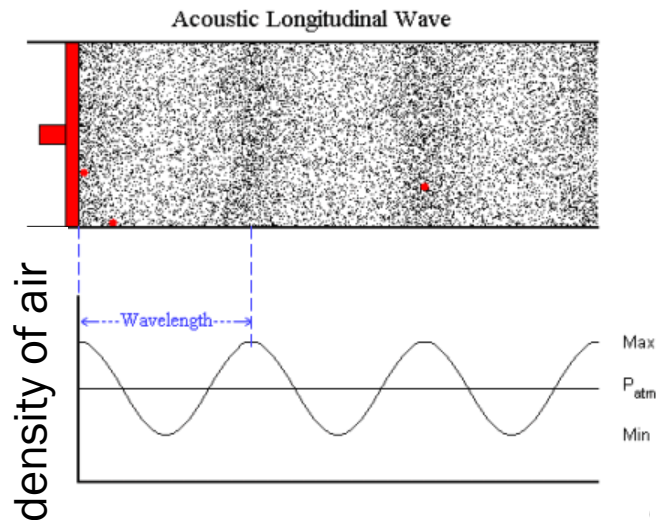
$$\text{Speed } u = \lambda/T = \lambda\nu$$

“Amplitude”

Some value assigned for each point of the space

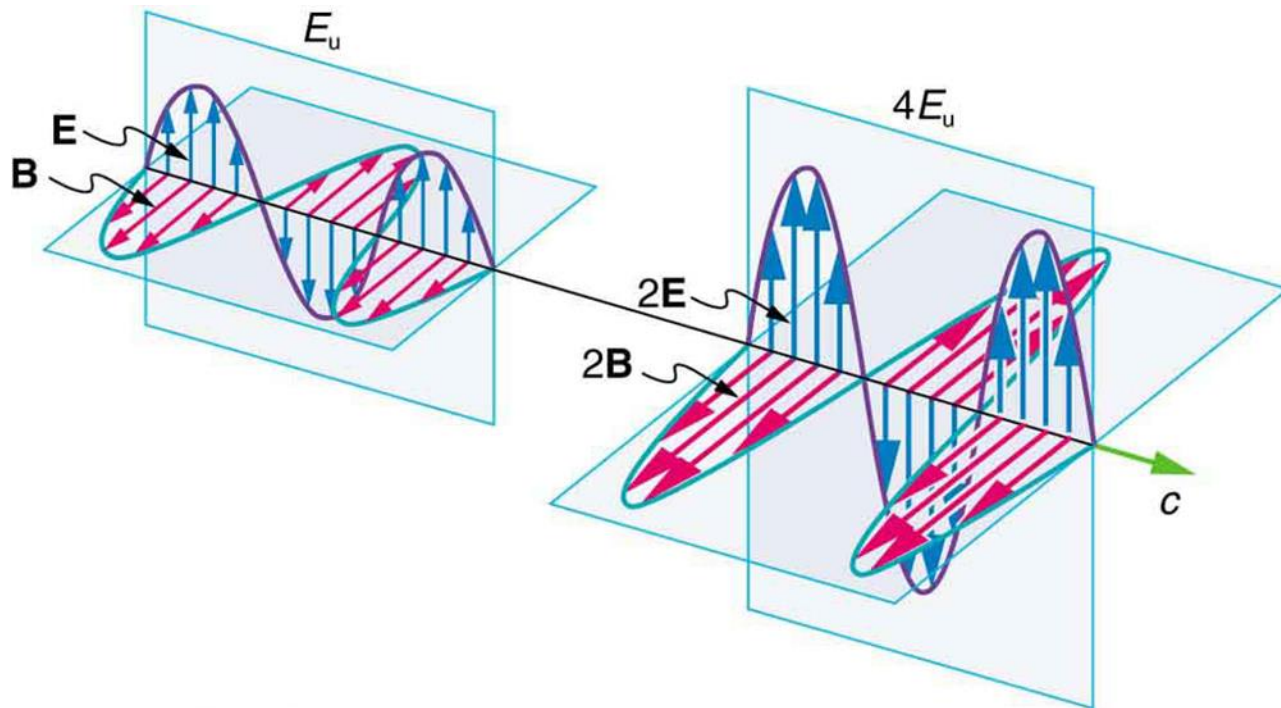


0 +1 +2 +1 0 -1 -2 -1 0 +1 +2 +1



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Light: Electromagnetic Wave

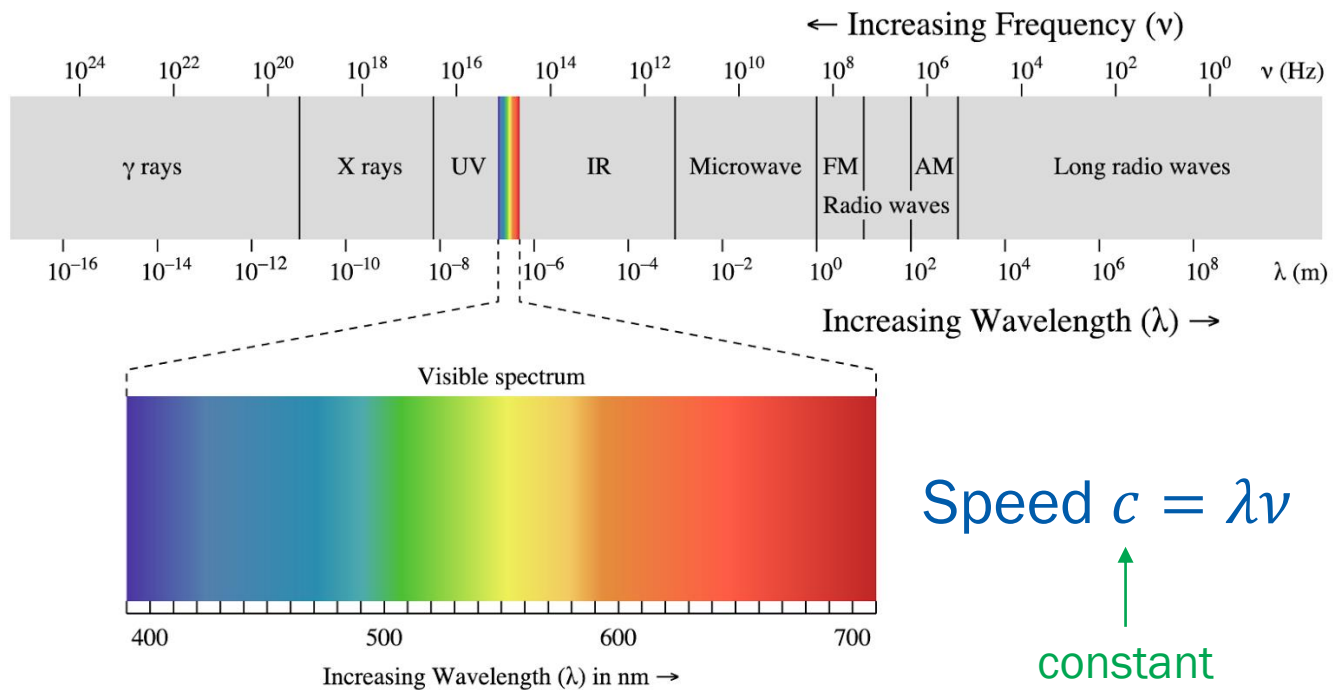


$$\text{Intensity } I_{\text{ave}} \propto E_0 B_0$$

(Image: <https://courses.lumenlearning.com/physics/chapter/24-4-energy-in-electromagnetic-waves/>)

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EM Wave Spectrum



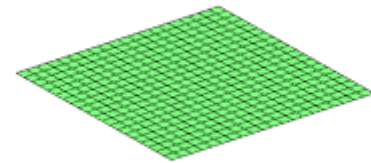
$$\text{Speed } c = \lambda \nu$$

$\uparrow$
constant
(Einstein)

(Image: <http://spmphysics.onlinetuition.com.my/2013/07/electromagnetic-waves-spectrum.html>)

Wave Equation

$$\frac{\partial^2 A}{\partial t^2} = u^2 \nabla^2 A$$



1D Traveling Wave



$$A(x, t) = A_0 \cos(kx - \omega t)$$

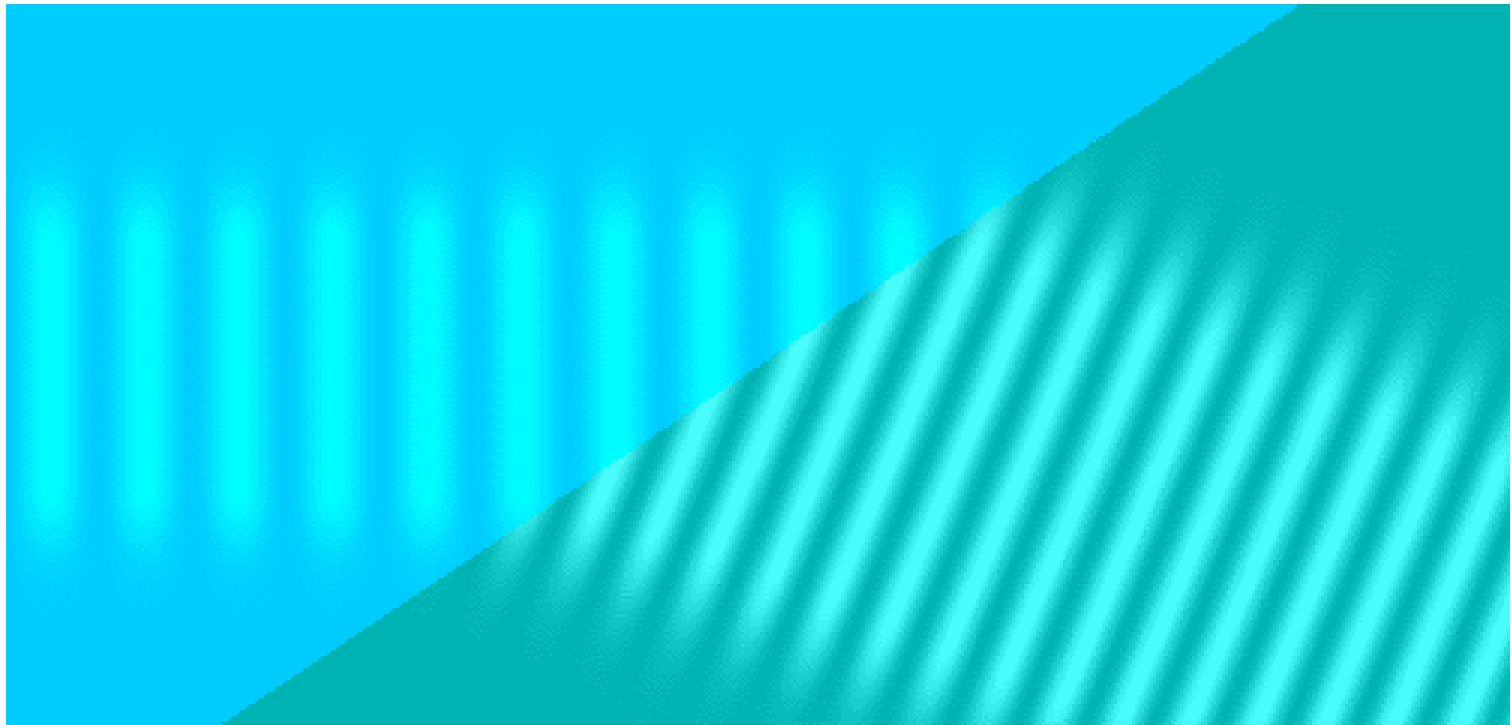
Wave equation: $\frac{\partial^2 A}{\partial t^2} = u^2 \frac{\partial^2 A}{\partial x^2}$

Substitution leads to

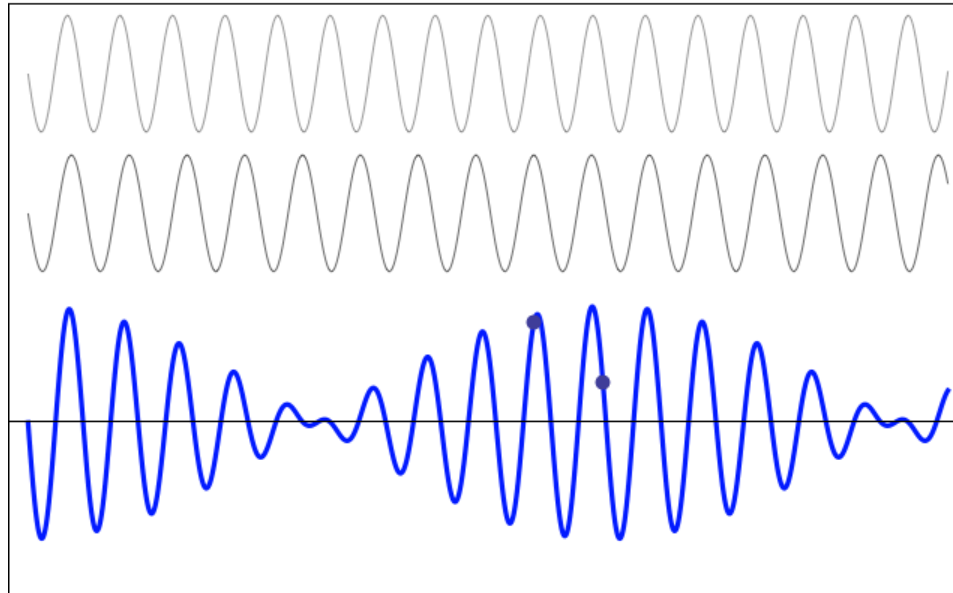
$$-A_0 \omega^2 \cos(kx - \omega t) = -A_0 u^2 k^2 \cos(kx - \omega t)$$

Hence, the wave equation is **satisfied** with $u = \omega/k$

Refraction

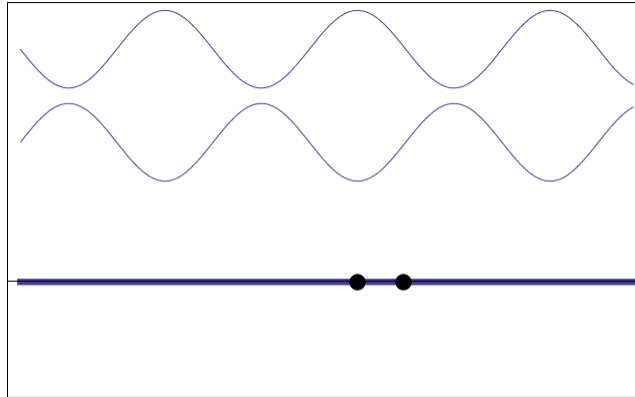


Superposition of Waves



$$A(x, t) = A_1 \cos(k_1 x - \omega_1 t) + A_2 \cos(k_2 x - \omega_2 t) + \dots$$

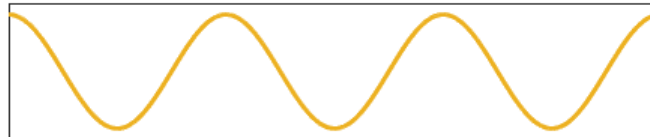
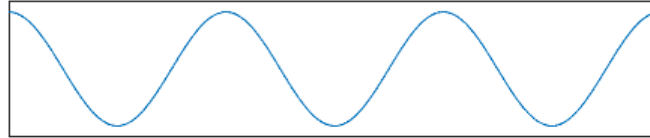
Standing Waves



$$\begin{aligned} A(x, t) &= A_0 \cos(kx - \omega t) + A_0 \cos(kx + \omega t) \\ &= A_0 [\cos(kx) \cos(\omega t) + \sin(kx) \sin(\omega t)] \\ &\quad + A_0 [\cos(kx) \cos(\omega t) - \sin(kx) \sin(\omega t)] \\ &= 2A_0 \cos(kx) \cos(\omega t) \end{aligned}$$

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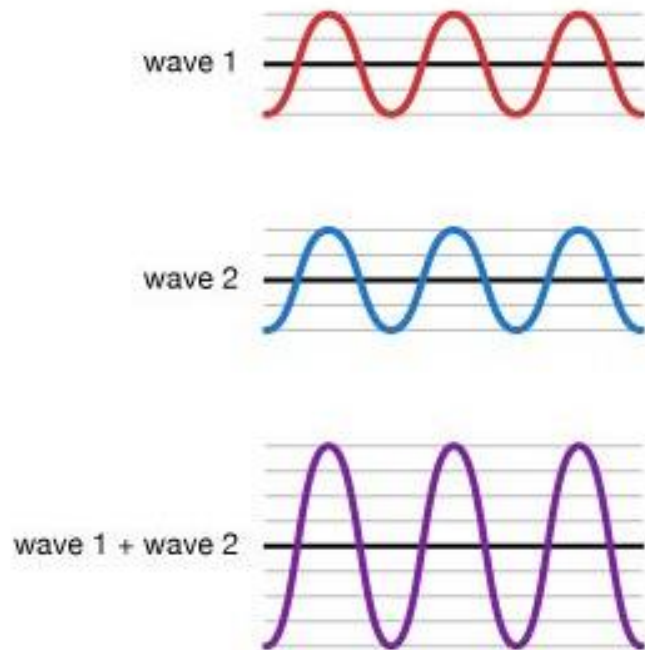
Standing Waves to Traveling Wave



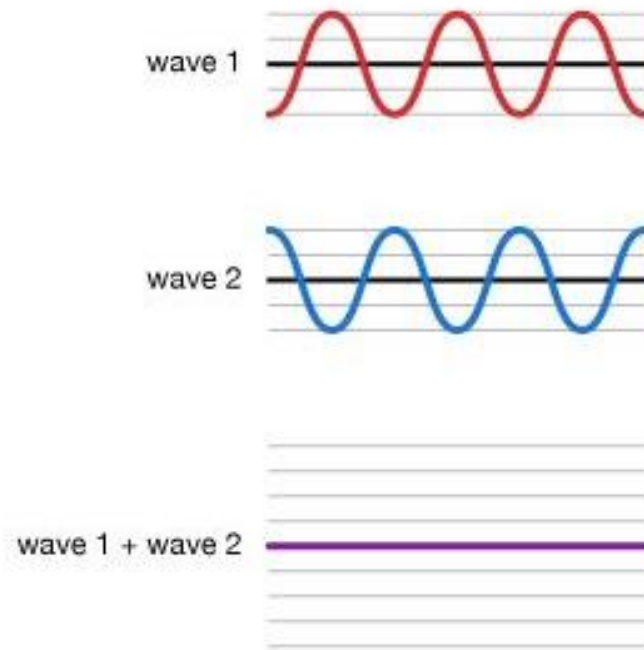
$$\begin{aligned} A(x, t) &= A_0 \cos(kx) \cos(\omega t) + A_0 \cos\left(kx - \frac{\pi}{2}\right) \cos\left(\omega t - \frac{\pi}{2}\right) \\ &= A_0 \cos(kx) \cos(\omega t) + A_0 \sin(kx) \sin(\omega t) \\ &= A_0 \cos(kx - \omega t) \end{aligned}$$

Interference

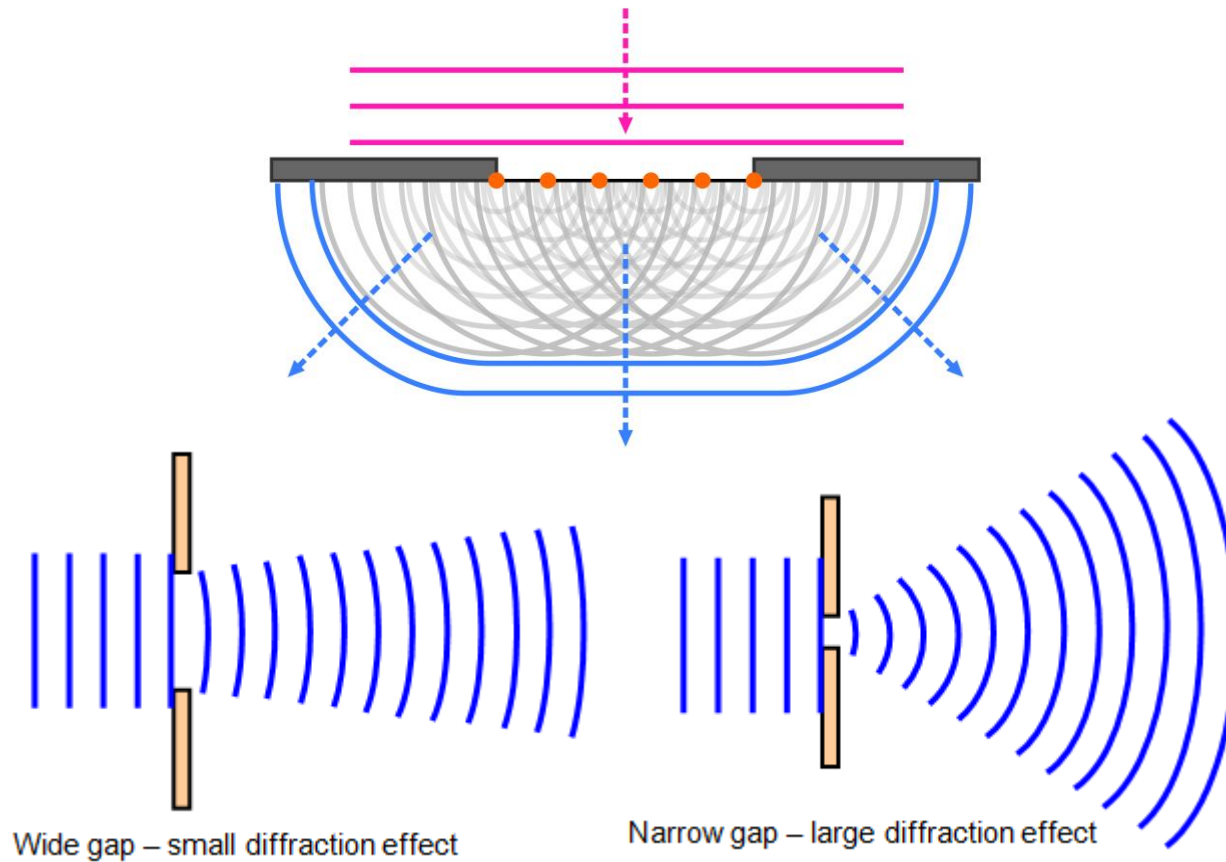
Constructive interference



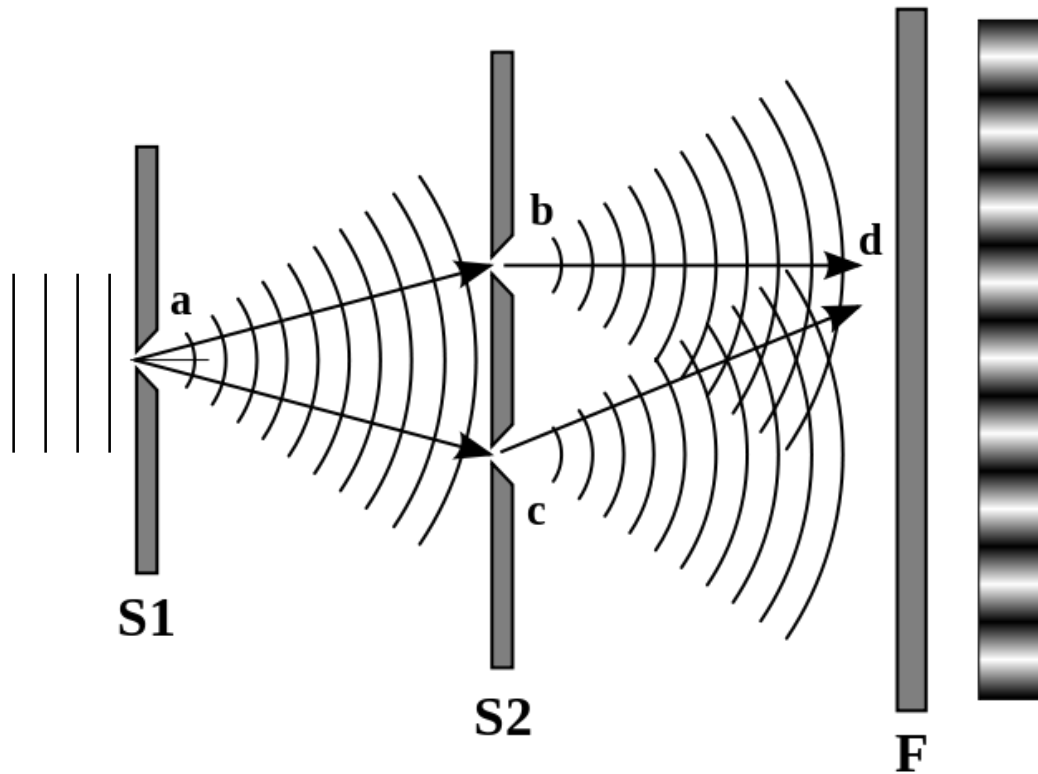
Destructive interference



Diffraction



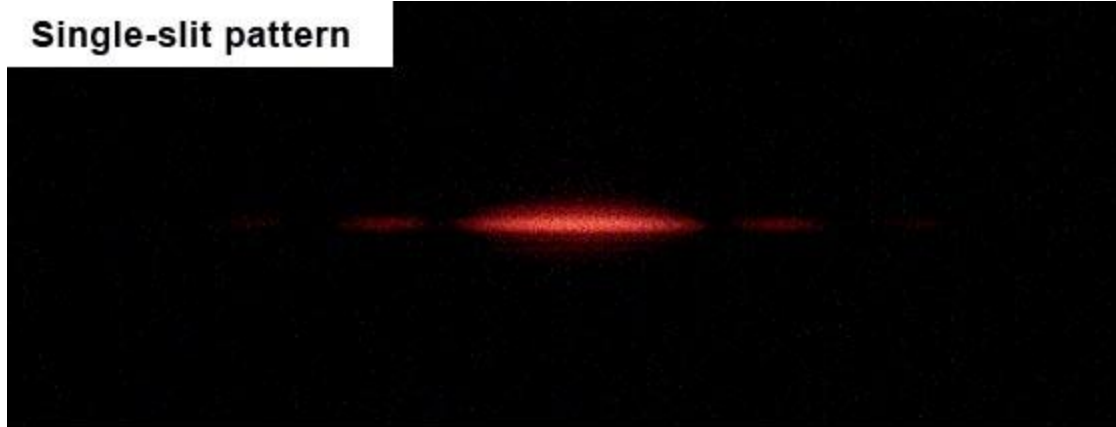
Double-Slit Experiment



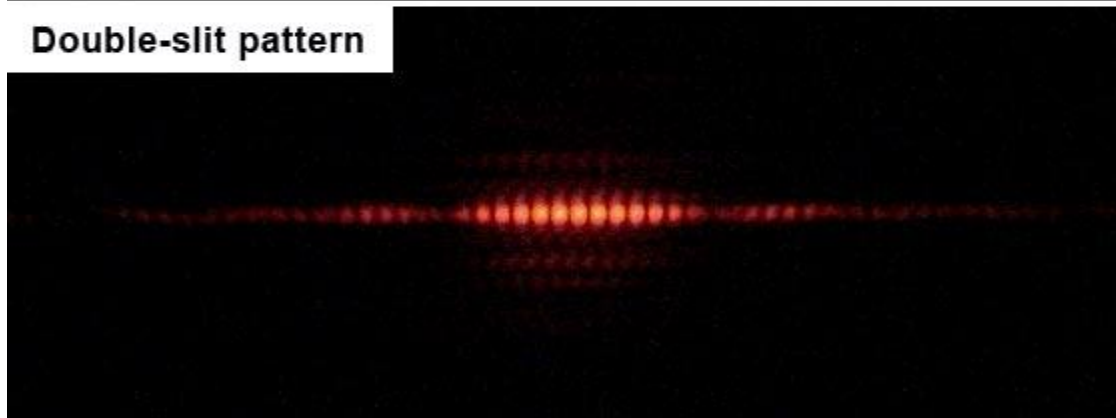
(Image: <https://commons.wikimedia.org/wiki/File:Ebohr1.svg>)

Double-Slit Experiment

Single-slit pattern



Double-slit pattern

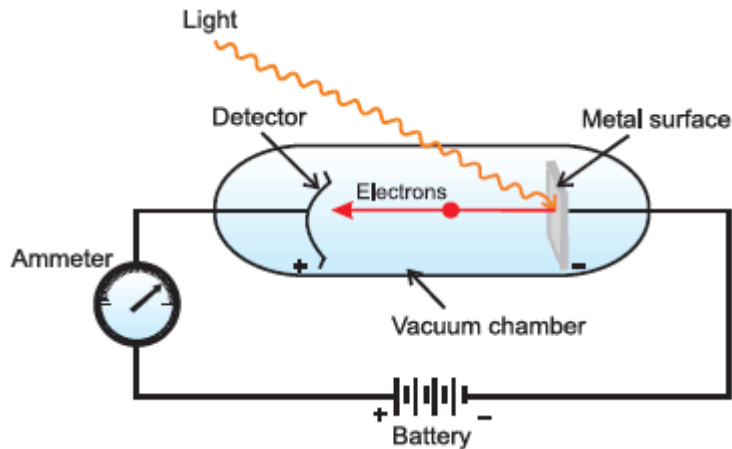


(Image: https://commons.wikimedia.org/wiki/File:Single_slit_and_double_slit2.jpg)



Yoan Choi
(18 months old)

Photoelectric Effect



- Intensity of light: $I \propto E_0 B_0$
- Frequency of light: $\nu = c/\lambda$

- “Threshold frequency” ν_{thres}
 - $\nu > \nu_{\text{thres}}$: electrons emitted; number of electrons $\propto$ intensity
 - $\nu < \nu_{\text{thres}}$: electrons not emitted, regardless of light intensity

Quantum Hypothesis

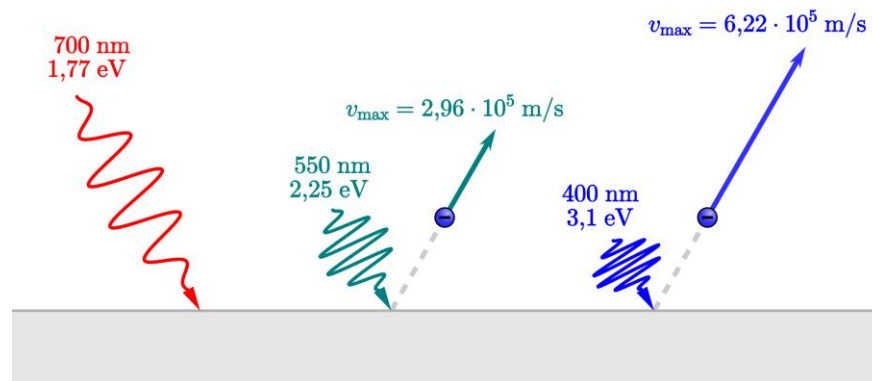
- Light can behave like a particle (“photon”)
- Energy of a photon

$$E = h\nu = hc/\lambda$$



Planck's constant

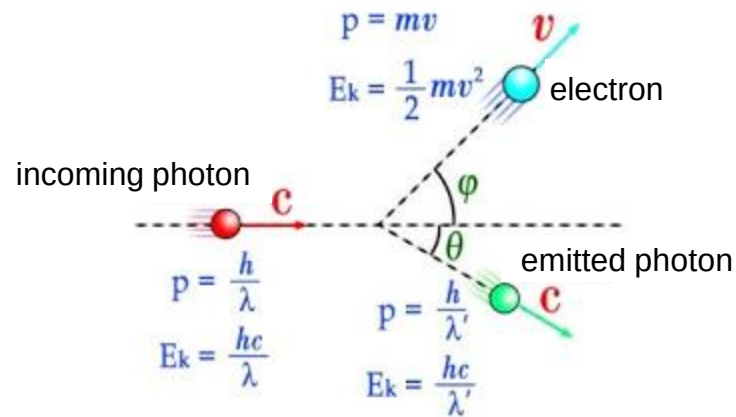
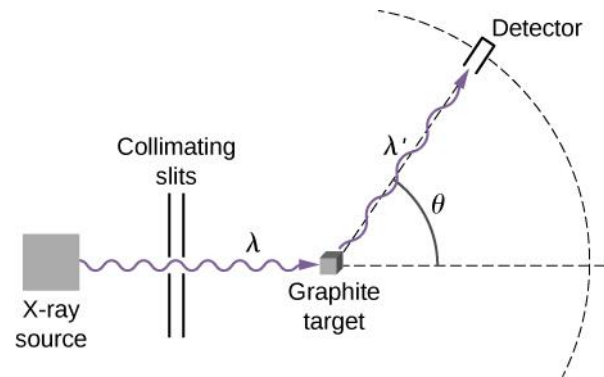
$$6.63 \times 10^{-34} \text{ J}\cdot\text{s}$$



(Image: <https://commons.m.wikimedia.org/wiki/File:Photoelectric-effect.svg>)

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Compton Scattering



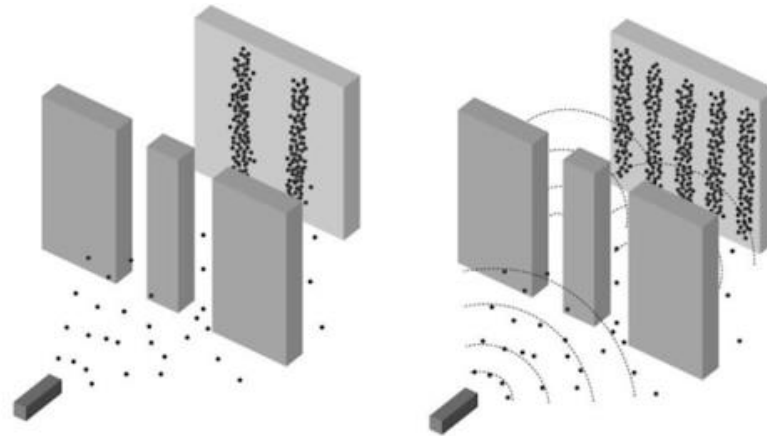
(Image: <https://opentextbc.ca/universityphysicsv3openstax/chapter/the-compton-effect/>
<https://blog.naver.com/toshizo/220528864147>)

Matter Wave

- Matter can behave like a wave (“matter wave”)
- Wavelength of a matter wave

$$\lambda = h/(mv)$$

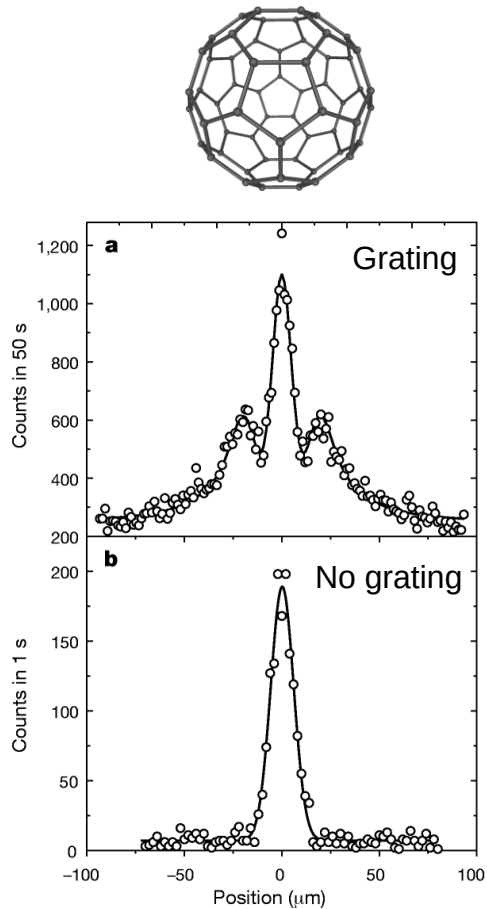
- Double-slit experiment



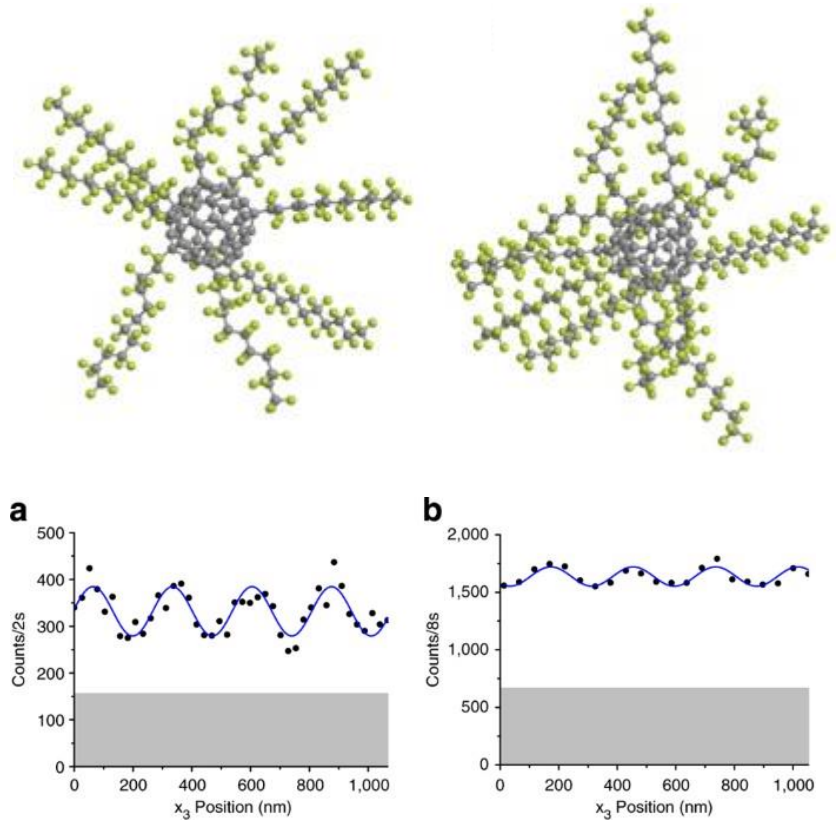
(Image: <https://blog.naver.com/msnayana/221374970924>)

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Diffraction of Large Molecules



Arndt et al., *Nature* (1999)



Gerlich et al., *Nat. Comm.* (2011)

Wave-Particle Duality

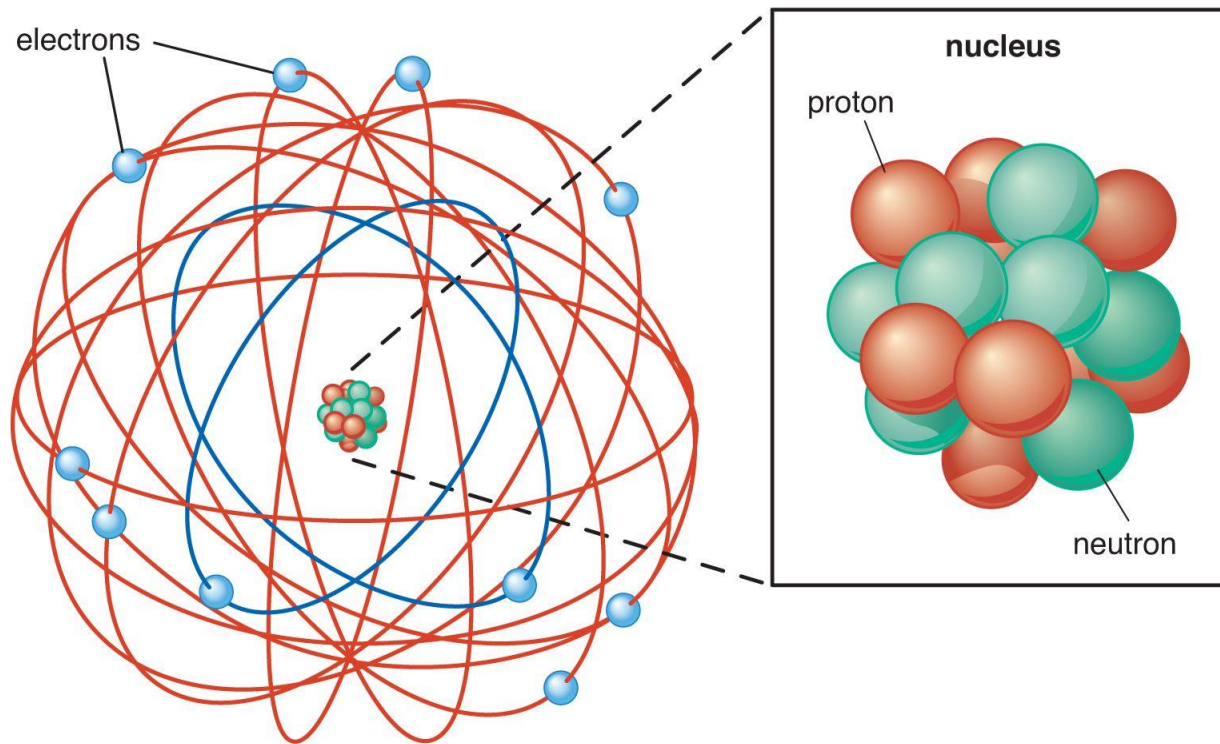
Light and matter exhibit

both properties of waves and particles

- Bohr's complementarity principle
 - Quantum objects can show wave or particle aspects depending on experiment
 - However, it is impossible to observe both the wave and particle aspects simultaneously

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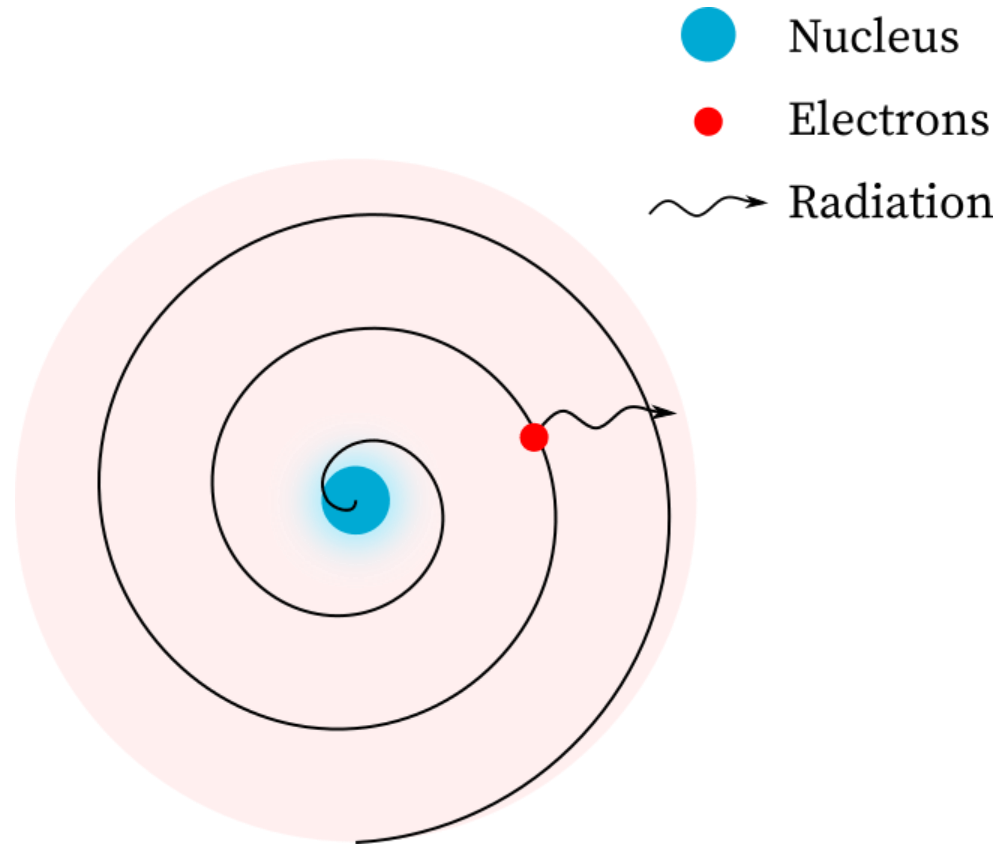
Rutherford Atomic Model



(Image: <https://www.britannica.com/science/Rutherford-model>)

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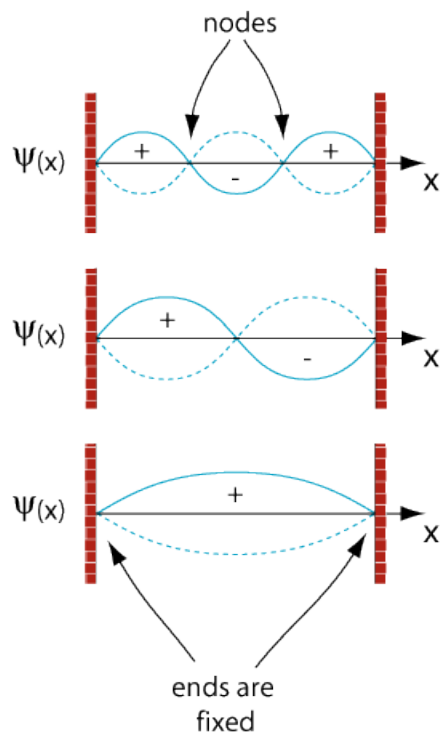
... and Its Problem



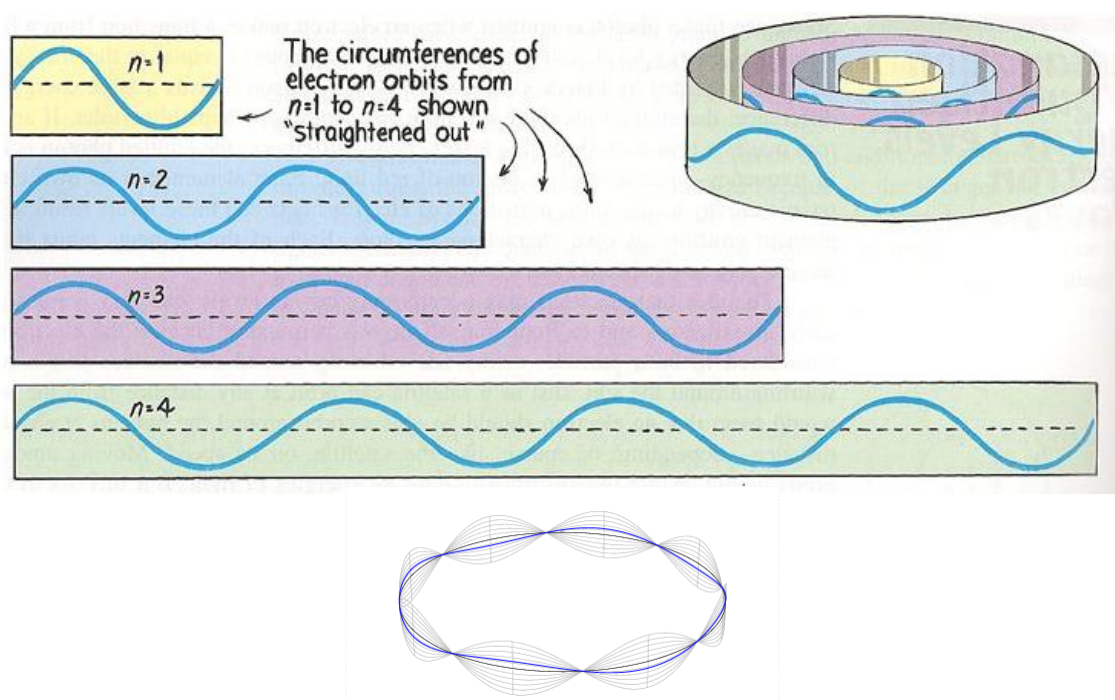
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Electrons as Waves

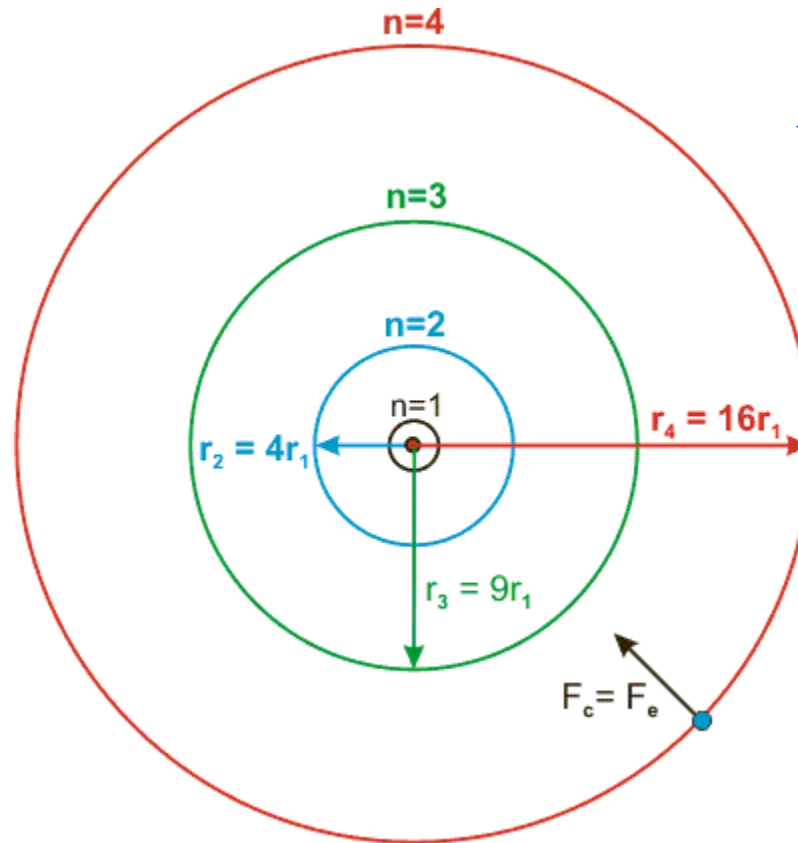
Standing wave



"Wave" of an electron



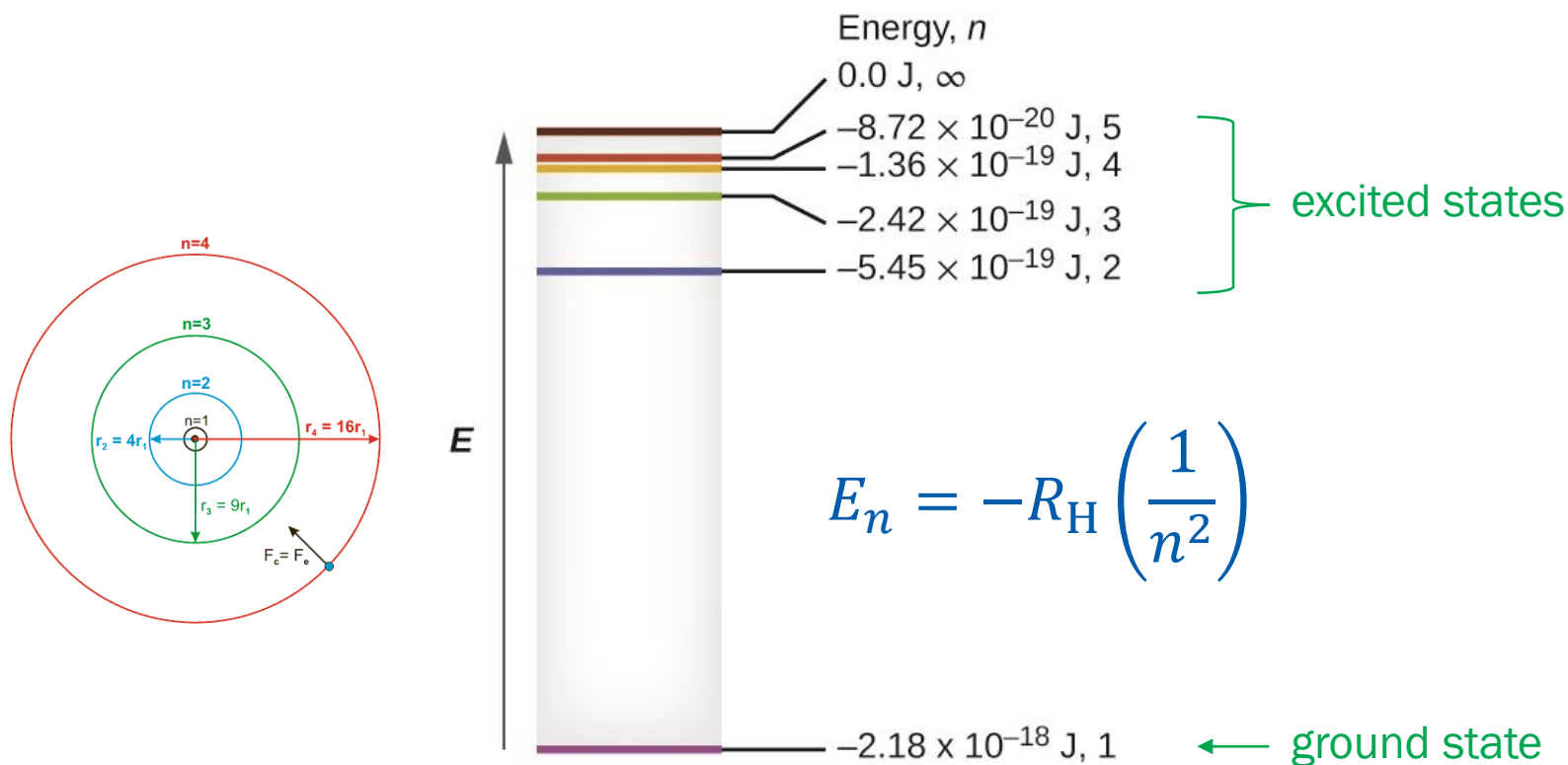
Bohr Atomic Model



$$E_n = -R_H \left(\frac{1}{n^2} \right)$$

Rydberg constant
 $2.18 \times 10^{-18} \text{ J}$

Energy Levels

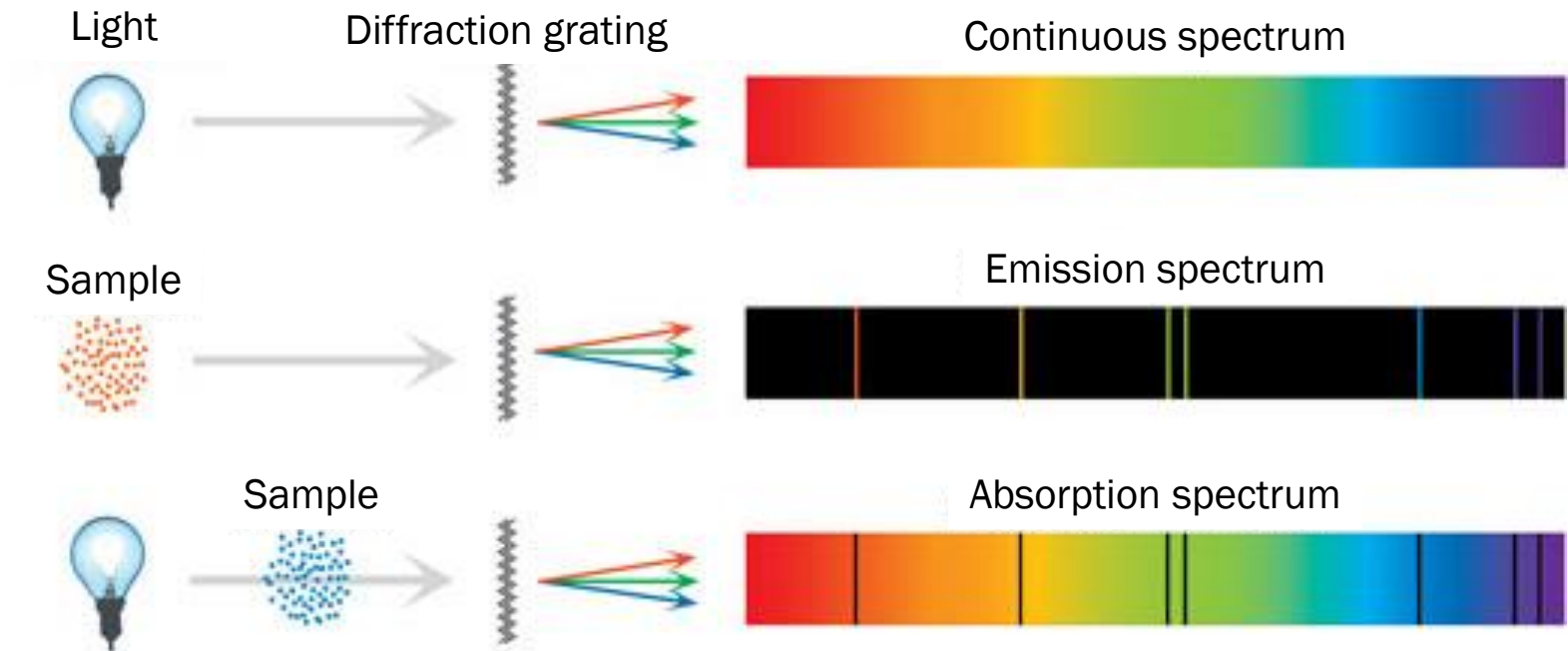


“Quantized” system!

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Spectroscopy

Study of the absorption and emission of **light** by **matter**



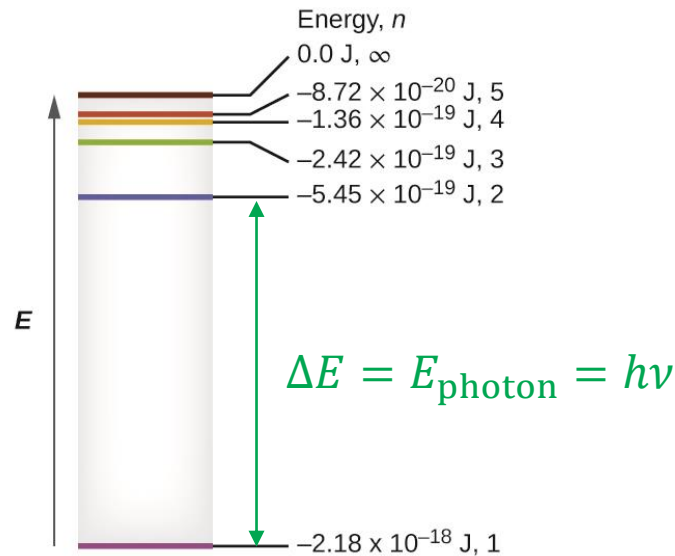
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Why Do We Get “Lines”?

- Quantization of light

$$E = h\nu$$

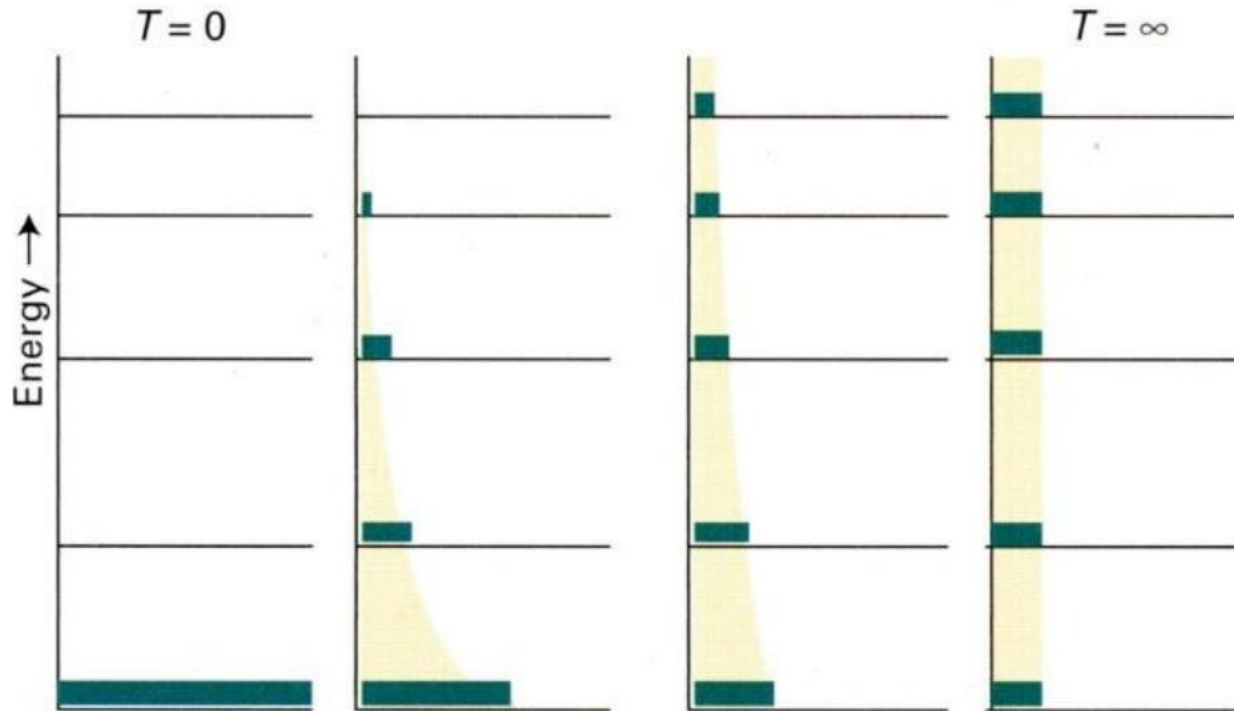
- Quantization of atomic energy



(Image: <https://opentextbc.ca/chemistry/chapter/6-2-the-bohr-model/>)

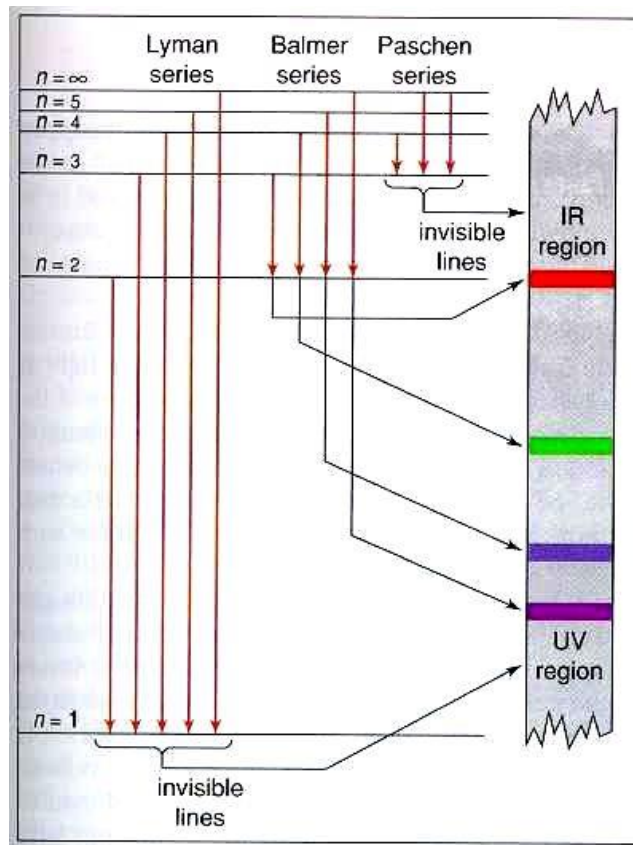
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Thermal Excitation



$$P(E) \propto e^{-E/k_B T} \text{ (Boltzmann factor)}$$

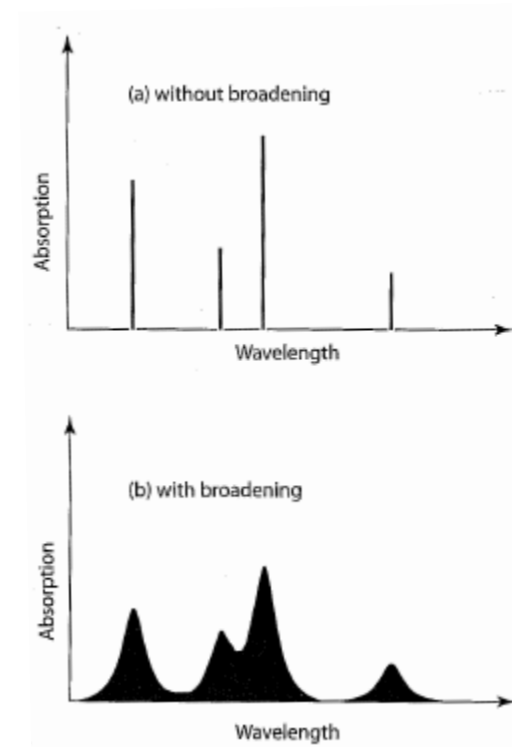
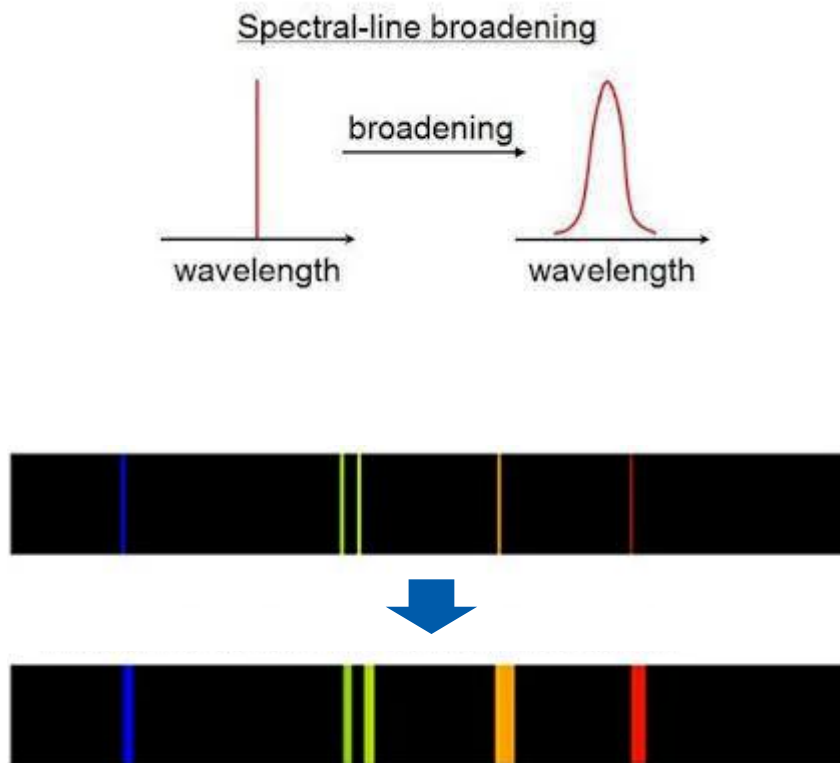
Transitions in H Atom



(Image: <https://bramblechemistry.weebly.com/4e2-bohr-model.html>)

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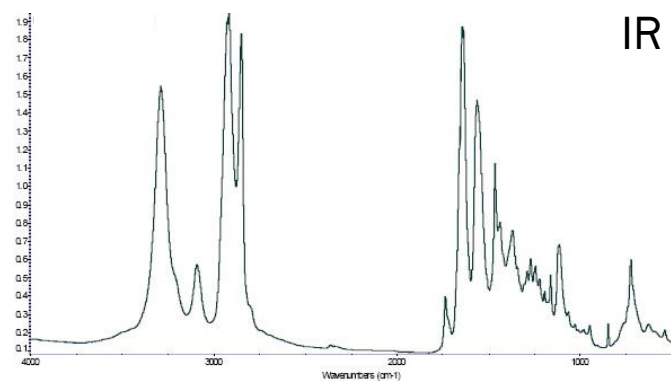
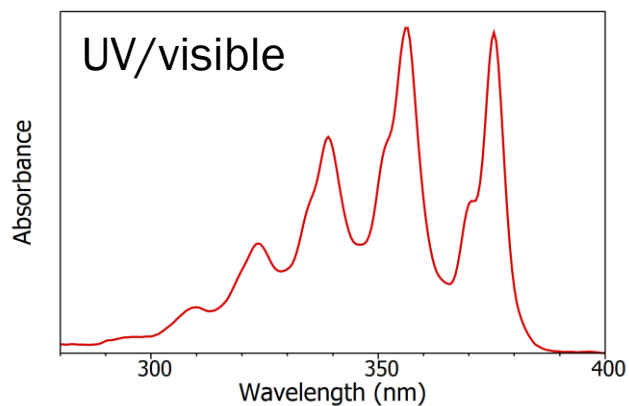
They Are NOT Lines Though...



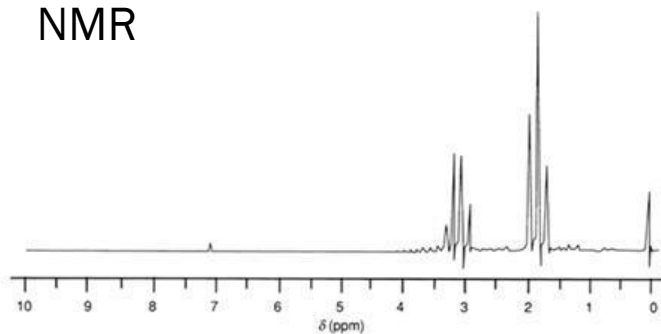
(Image: https://uomustansiriyah.edu.iq/media/lectures/6/6_2018_12_22!12_29_47_AM.pdf
<https://skyfallmeteorites.com/education-research/glossary/thermal-doppler-broadening/>)

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Real Spectra



NMR



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Breakdown of the Bohr Model

- The Bohr model can explain the line spectrum of the hydrogen atom, but ...
- It cannot explain other atoms
- It cannot explain the intensity of lines
- It cannot explain the hydrogen atom under the magnetic field

We need a new theory: modern quantum mechanics

